SEVA FIRST INNOVATION CHALLENGE 2026 * GOVTECH TRACK

CANDIDATE: SF-2026-MORM

SEVA-PARAKH

by MorMati Labs Solo Student Innovator - B.Tech CSE, NIT Raipur

A vernacular pre-submission document audit and rejection-decoder tool that helps **citizens and operators catch application defects BEFORE portal upload.**

DETERMINISTIC RULE ENGINE

100% Policy Grounded (Zero AI Hallucination)

Validates dates, circulars & spelling via regex & GR index.

REGIONAL SLM & VERNACULAR AUDIO

Quantized Edge Model (Chhattisgarhi & Marathi)

Translates cryptic administrative error codes into spoken advice.

PROBLEM STATEMENT

The Problem: Preventable Rejection Overhead

Applications fail DBT & scholarships not due to ineligibility, but due to preventable document defects.

WHY APPLICATIONS GET "SENT BACK"

Thousands of scholarship and welfare applications fail scrutiny due to clerical errors: expired income certificates, name spelling variances (e.g., Sahu vs Saw), and unlinked NPCI bank accounts.

THE 100-CASE SYNTHETIC TECHNICAL BENCHMARK

In our benchmark evaluation across 100 synthetic government dossiers, standard e-District and DBT portals blindly accept defective PDFs at upload, **shifting 100% of the verification burden onto human scrutiny clerks.**

THE EXISTING AI FALLACY

Chatbots only recite scheme rules. They CANNOT cross-audit physical scanned documents or verify certificate issue dates against circulars.

30%+

Govt Scrutiny Clerk Overhead

Clerks re-open & reject preventable clerical defects.

Rs. 150 - Rs. 300

Cash Burn Per Citizen Per Rejection

Lost in multiple bus trips, re-scanning, and wages.

12 - 15 Mins

Wasted Per Operator Per Defect

Manual document squinting creates center backlogs.

Beneficiaries: Rural students & Aaple Sarkar / Lok Seva Kendra VLEs | States: Maharashtra & Chhattisgarh

CORE INNOVATION

DETERMINISTIC RULES + REGIONAL SLM

Our Solution: "Rules Decide. AI Explains."

1 **Drop & On-Device Redact**

Client-side OCR extracts text and auto-masks 12-digit Aadhaar.

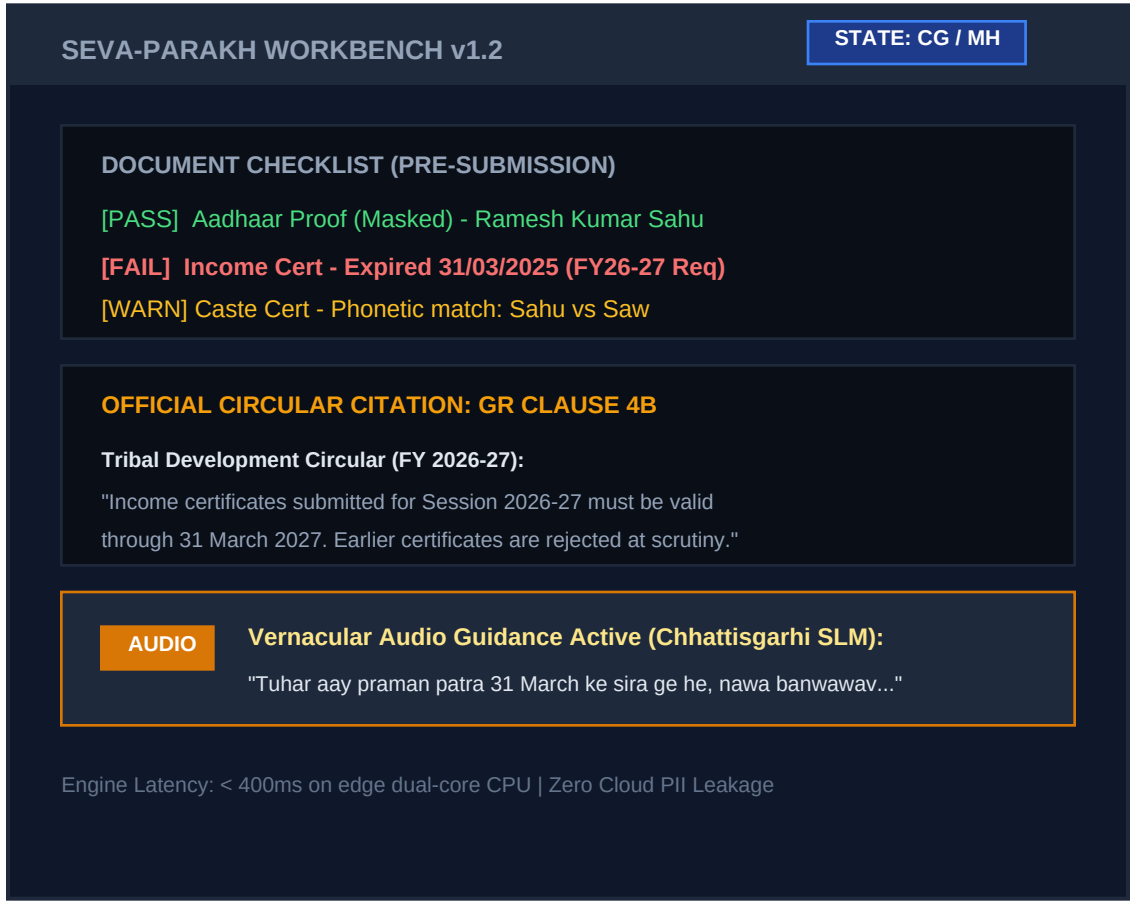
2 **Deterministic Rule Engine (0% Hallucination)**

Validates expiry dates against State GR circulars & cross-script spelling.

3 **Vernacular Audio Guidance**

Quantized SLM translates rejections into spoken Chhattisgarhi/Marathi.

METRIC	STANDARD FLOW	SEVA-PARAKH WORKBENCH
Validation Cost	Rs. 150+ lost/trip	**< Rs. 0.15 / audit**
Check Time	12-15 mins manual	**< 4 seconds automated**
Citizen Next Step	Cryptic code	**Spoken Dialect Audio**



Key USP: 100% Deterministic policy validation coupled with quantized Indic dialect fine-tunes.

IMPLEMENTATION & BUDGET

WORKING PROTOTYPE VERIFIED

Feasibility & Grounded Unit Economics

01 / TECH STACK

Edge Production Stack

Client-Side OCR:

PaddleOCR / Tesseract WASM with on-device 12-digit Aadhaar masking.

Deterministic Engine:

Python regex + date arithmetic; official GR policy master circulars.

Regional SLM:

Quantized Indic-Mistral/Llama3 with Bhashini TTS/ASR voice pipelines.

Hardware: Low-spec dual-core PCs.

02 / UNIT ECONOMICS

Rs. 9.5L Pilot Budget

Setup Cost per Center

Rs. 800 - Rs. 1,000

Monthly Run Cost / Kendra

~ Rs. 200 / month

50-CENTER PILOT (6 MOS)

Rs. 9.50 Lakhs Total

Fits Rs. 10L Work Order Cap

Zero expensive GPU cloud required.

03 / 12-MONTH TIMELINE

Execution Milestones

Months 01 - 03: MahaDBT Rules

Finalize Maharashtra scholarship circulars; 10-center Aaple Sarkar sandbox trial.

Months 04 - 06: 50-Center Pilot

Deploy across CG/MH border district belt; target 25,000 real verifications.

Months 07 - 12: State Scale-Up

API integration with State IT portals; prep for statewide public procurement.

Target: Statewide expansion.

TARGET GOV PARTNERS: *** MahaIT & Aaple Sarkar Network** *** MSInS / SEEID Incubation** *** Zilla Parishad Social Welfare** *** CSC Network**

Budget compliant with Seva First Innovation Challenge Rs. 10 Lakh Work Order guidelines.

IMPACT & SCALE

Systemic Impact & Scalability

75,000+

Beneficiaries Reached (Yr 1)

Across 50 pilot centers; preventing wrongful rejection of rural students.

88% - 94%

Prevented Defect Rate

Evaluated on 100 synthetic cases; catches validity & spelling issues.

1.5 Hours

Daily Time Saved Per VLE

Frees operators from manual checking; increases daily throughput by 25%.

SOCIAL EQUITY & LINGUISTIC INCLUSION

Standard portals disenfranchise non-English and non-standard Marathi speakers through cold SMS rejection notices with bureaucratic error codes.

Seva-Parakh bridges this digital chasm by translating scrutiny findings **into spoken, conversational regional dialects (Chhattisgarhi / Marathi).**

Result: Closes the digital divide for first-generation applicants.

STATE-AGNOSTIC MODULAR ARCHITECTURE

The core deterministic engine is fully separated from state-specific policies. Expanding to a new state requires only authoring a single JSON rule pack:

```
/seva_rules/mh_mahadbt.json  <->  /seva_rules/cg_edistrict.json
```

Ready for statewide rollout across 38,000+ Aaple Sarkar Kendras.

Zero recurring per-user cloud licensing fees. Fully sustainable on local VLE center processing.

Team, Traction & Rs. 10L Work Order Request

SOLO INNOVATOR

Jayesh Sahu

Lead Developer & Founder, MorMati Labs

B.Tech Computer Science & Engineering, NIT Raipur

Specialization: Quantized SLM deployment, Indic NLP, and deterministic GovTech validation. Built and benchmarked the Seva-Parakh engine.

Working prototype evaluated with 100-case synthetic validation.

FACULTY ADVISORY

Faculty Mentor - Dept. of CSE, NIT Raipur

Advising on systems security, zero-leakage PII architectures & data verification.

WHAT WE SEEK FROM SEVA FIRST CHALLENGE:

[X] Rs. 10 Lakh Work Order / Grant:

Fully covers edge calibration, training, and 50-center pilot support.

[X] 50-Center Pilot Mandate:

Official clearance to deploy workbench in select Aaple Sarkar Kendras.

[X] GovTech Scrutiny Circular Access:

Direct access to edge-case clerk rejection checklists to expand rules.

COMMITMENT TO THE SEVA FIRST JURY:

"The Rs. 10 Lakh Work Order will bridge our validated Chhattisgarhi prototype into a production-grade, 50-center MahaDBT Marathi deployment - eliminating preventable welfare rejections before portal upload."